

An Online Open-Loop S-Curve Velocity Profile Control Method for Stepping Motors on FPGA

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ABSTRACT

Generating S-curve profiles online is computationally intensive for microcontrollers. The authors present an online open-loop S-curve velocity profile control method implemented on FPGAs and high-speed processors, demonstrating smooth acceleration and reduced vibration in stepper motors.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Justifies the offloading of S-curve computation and the use of dedicated high-speed microcontrollers for trajectory planning.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Zhong Chen provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.